In[•]:=
```
ClearAll[t, a, rho, zKin, zGrav, zTotal, plotRedshift, plotDensity, combinedPlot];

(*1. Lock in the Static Parameters*)
lookbackMax = 13.9;
kinematicH = 0.0;
(*Set to zero to isolate the gravitational effect*)pastDensity = 8.0;
presentDensity = 0.5;
densityDecay = 0.3;
gravCoupling = 0.05;

(*2. Define the Functions*)
a[t_] := Exp[kinematicH * t];
rho[t_] := presentDensity +
   (pastDensity - presentDensity) * Exp[-densityDecay * (lookbackMax - t)];
zKin[t_] := a[lookbackMax] / a[lookbackMax - t] - 1;
zGrav[t_] :=
  gravCoupling * Integrate[rho[tau], {tau, lookbackMax - t, lookbackMax}];
zTotal[t_] := (1 + zKin[t]) * (1 + zGrav[t]) - 1;

(*3. Generate the Individual Plots*)
plotRedshift = Plot[{zKin[t], zGrav[t], zTotal[t]}, {t, 0, lookbackMax},
   PlotStyle → {Directive[Thick, Black, Dashed], Directive[Thick, GrayLevel[0.4],
      Dotted], Directive[Thick, Black]}, PlotRange → All, Frame → True,
   FrameLabel → {"Lookback Distance (BLY)", "Observed Redshift (z)"},
   PlotLegends → {"z (Kinematic)", "z (Gravitational Lag)", "z (Total Observed)"},
   PlotLabel → "Compound Redshift: Decoupling Physical Expansion from Density Lag",
   ImageSize → 600];

plotDensity = Plot[rho[t], {t, 0, lookbackMax},
   PlotStyle → Directive[Thick, Black], PlotRange → {0, pastDensity + 1},
   Filling → Axis, FillingStyle → Directive[Gray, Opacity[0.2]], Frame → True,
   FrameLabel → {"Lookback Distance (BLY)", "Effective Mass Density of Shells"},
   PlotLabel → "Pound-Rebka Asymmetry: The Retarded Density Drop-Off",
   ImageSize → 600];

(*4. Combine and Export as a Vector PDF*)
combinedPlot = GraphicsColumn[{plotRedshift, plotDensity}, ImageSize → 650];

Export["pound_rebka_asymmetry.pdf", combinedPlot, "PDF"]
```
Out[•]=
```
pound_rebka_asymmetry.pdf
```
In[•]:=
```
SystemOpen["pound_rebka_asymmetry.pdf"]
```